NOTE Swashbuckle uses the old Swagger nomenclature rather than OpenAPI in its method names. You should think of OpenAPI as the name of the specification and Swagger as the name of the tooling related to OpenAPI, as described in this post: http://mng.bz/o18M.

Listing 11.1 Adding OpenAPI support to a minimal API app using Swashbuckle

```
using System.Collections.Concurrent;

WebApplicationBuilder builder = WebApplication.CreateBuilder(args);

builder.Services.AddEndpointsApiExplorer();     ❶
builder.Services.AddSwaggerGen();               ❷

WebApplication app = builder.Build();

var _fruit = new ConcurrentDictionary<string, Fruit>();

app.UseSwagger();                               ❸
app.UseSwaggerUI();                             ❹

app.MapGet("/fruit/{id}", (string id) =>
    _fruit.TryGetValue(id, out var fruit)
        ? TypedResults.Ok(fruit)
        : Results.Problem(statusCode: 404));

app.MapPost("/fruit/{id}", (string id, Fruit fruit) =>
    _fruit.TryAdd(id, fruit)
        ? TypedResults.Created($"/fruit/{id}", fruit)
        : Results.ValidationProblem(new Dictionary<string, string[]>
          {
              { "id", new[] { "A fruit with this id already exists" } }
        }));

app.Run();
record Fruit(string Name, int Stock);
```

❶ Adds the endpoint-discovery features of ASP.NET Core that Swashbuckle requires

❷ Adds the Swashbuckle services required for creating OpenApi Documents

❸ Adds middleware to expose the OpenAPI document for your app

❹ Adds middleware that serves the Swagger UI

With the changes in this listing, your application exposes an OpenAPI description of its endpoints. If you run the app and navigate to /swagger/v1/swagger.json,